

From Research to Real-World Applications: AI and NLP in Industry

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Outlines

1. Introduction to Factored Machine Translation
2. Research Objectives and Key Contributions
3. Research Methodology
4. Preprocessing: Part-of-Speech (POS) Tagging
5. Experiments and Results
6. Summary of Research Work
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Introduction

What is Machine Translation (MT)?

- NLP task that automatically translates text from a **source language** to a **target language**.

Statistical Machine Translation (SMT)

- A data-driven approach that learns **statistical relationships** between source and target languages from parallel corpus.
- Translates text by selecting the **most probable words or phrases** using translation and language models.
- Effective for many language pairs but has limitations in capturing **long-range context** and producing fluent translations.

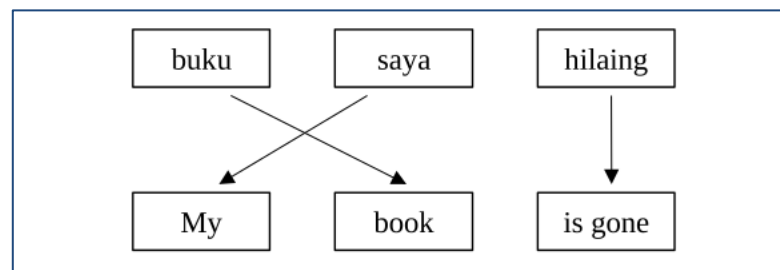


Figure 1.1 Standard Phrase translation model

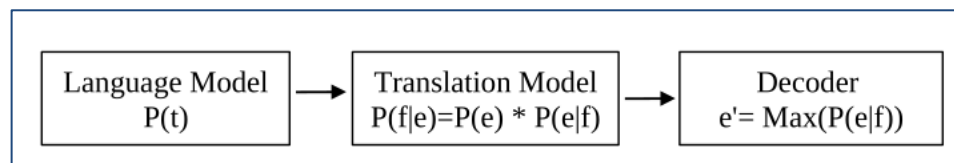


Figure 1.2 Standard Translation model component

Introduction

Neural Machine Translation (NMT)

- A deep learning-based approach that performs **end-to-end translation** using an **Encoder-Decoder (Seq2Seq)** architecture.
- Learns the **context and semantic meaning** of the entire sentence using an **Attention Mechanism**.
- Produces more **accurate, fluent, and context-aware** translations than traditional SMT and is the current state-of-the-art approach.

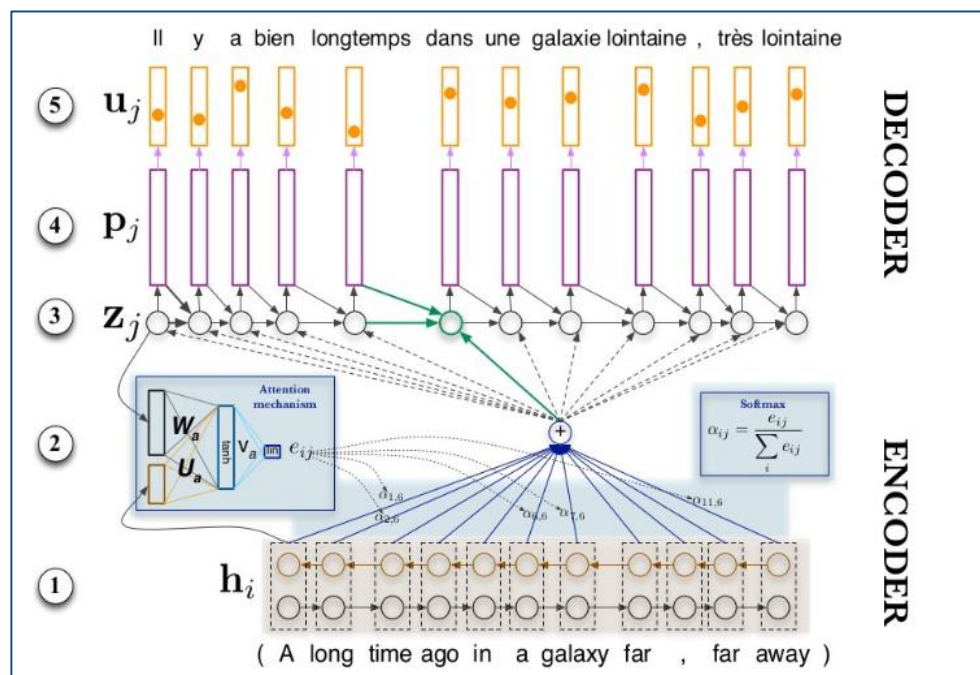


Figure 2.1 Attention-based NMT system

Introduction

What is Factored Translation?

- Factored Translation was first introduced in Statistical Machine Translation (SMT) as an extension of Phrase-Based SMT (PB-SMT).
- Instead of using only the surface form (words), it incorporates additional linguistic features (factors) to improve translation quality.

Common Linguistic Factors

- ❖ **Surface Form** – original word
- ❖ **Lemma** – dictionary/base form
- ❖ **Part-of-Speech (POS)** – grammatical category
- ❖ **Morphological Tags** – grammatical features (e.g., tense, number, gender)

Why Use Factored Translation?

- ❖ Reduces **data sparsity**
- ❖ Improves phrase and language model estimation
- ❖ Particularly effective for **morphologically rich** and **under-resourced languages**

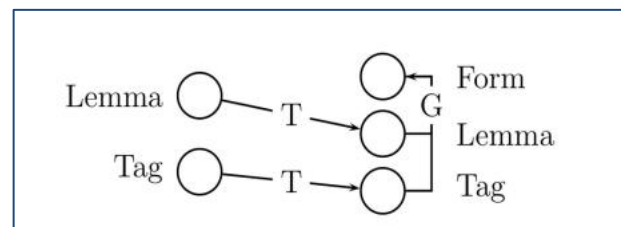


Figure 2.2 An example of translation and generation steps

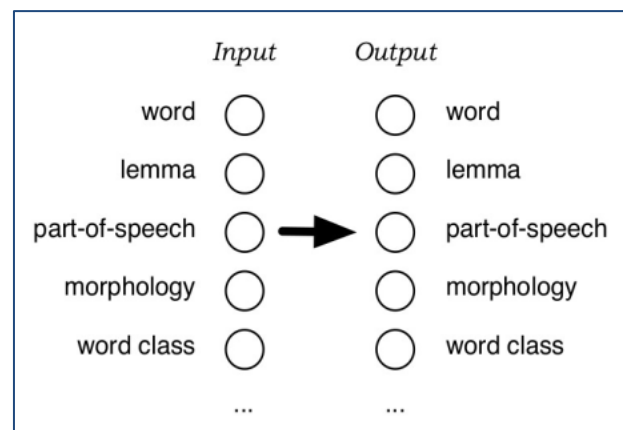


Figure 2.3 Factored representation of input and output words

Introduction

Factored Neural Machine Translation (FNMT)

- Extension of the standard **Neural Machine Translation (NMT)** architecture.
- Unlike conventional NMT, **Factored NMT generates multiple output factors simultaneously**, rather than only predicting the target word.

Advantages of Factored NMT

- ❖ Reduces **out-of-vocabulary (OOV)** problems
- ❖ Handles **large vocabulary** more effectively
- ❖ Mitigates **data sparsity** and **morphological complexity**
- ❖ Generates new word forms by combining **lemmas** with **linguistic features**
- ❖ Improves overall **translation quality and performance**

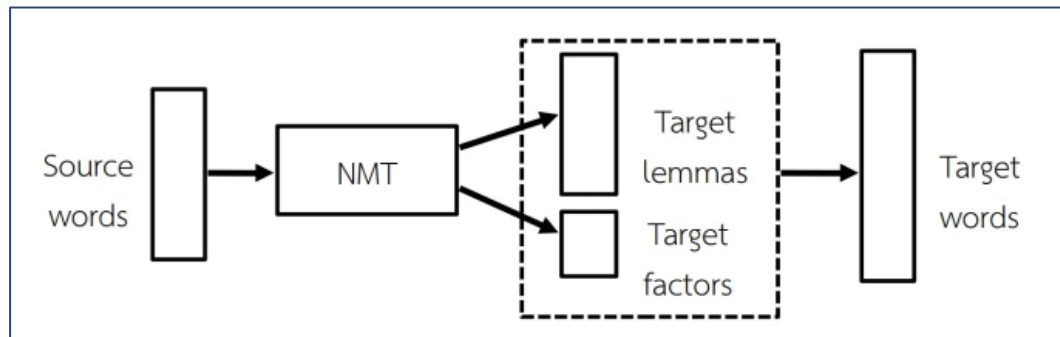


Figure 3.1 NMT system pipeline with factored output

Introduction

Motivation of Proposed System

- Motivated by various benefits of **factored MT**, most researchers have focused on improving translation performance for **morphologically rich and under-resourced languages**.
- Annotating the source side, target side, or both sides with linguistic features can improve the translation quality of machine translation.
- There have been no studies on effectively incorporating linguistic features into NMT models for Thai–English and Thai–Myanmar low-resource language pairs.



Introduction

Motivation of Proposed System

- **Transformer model** outperforms **RNN and CNN models**, particularly in translation tasks.
- Because it allows parallelization to alleviate the problem of learning long-term memory dependency and uses **self-attention** to build dependencies between inputs and outputs.
- Therefore, we propose **Transformer models, multi-source Transformer models, and shared-multi-source Transformer models** with additional linguistic features for low-resource language pairs.



Introduction

Research Hypothesis

- Applying POS tag information on [multi-source transformer](#) and [shared-multi-source transformer](#) architectures had to be better than using only word vectors



Introduction

Research Questions

1. How much could NMT performance increase from adding POS-tag information to transformer neural network architecture?
2. How much does NMT with POS-tag information affect low-resource language pairs?





Research Objectives and Contributions

Research Objectives

- To apply additional linguistic features to NMT models for improving the translation performance of under-resourced languages such as Thai and Myanmar
- To alleviate the data sparseness and out-of-vocabulary word problems for under-resourced languages
- To enhance the translation performance of under-resourced languages

Contributions

1. Building Myanmar POS tagger to be used in factored machine translation.
2. Developing a Thai POS tagger for POS tagging of Thai text data.
3. Proving that incorporating POS tag information on a transformer architecture can improve NMT performance for low-resource language pairs.
4. Proving that incorporating POS tag information on multi-source and shared-multi-source transformer architectures can dramatically improve its NMT performance over baseline transformer architecture.

Research Gap

- Most of FMT systems use linguistic features only on source or target side, but didn't use both sides
- There was no study on incorporating linguistic features for NMT models for Thai–English and Thai–Myanmar
- We added linguistic information such as POS tags or UPOS tags to each word on source side or target side, and both sides
- To apply the proposed models, we used transformer-based models:
 - transformer
 - multi-source transformer
 - shared-multi-source transformer



Research Methodology

Transformer Model

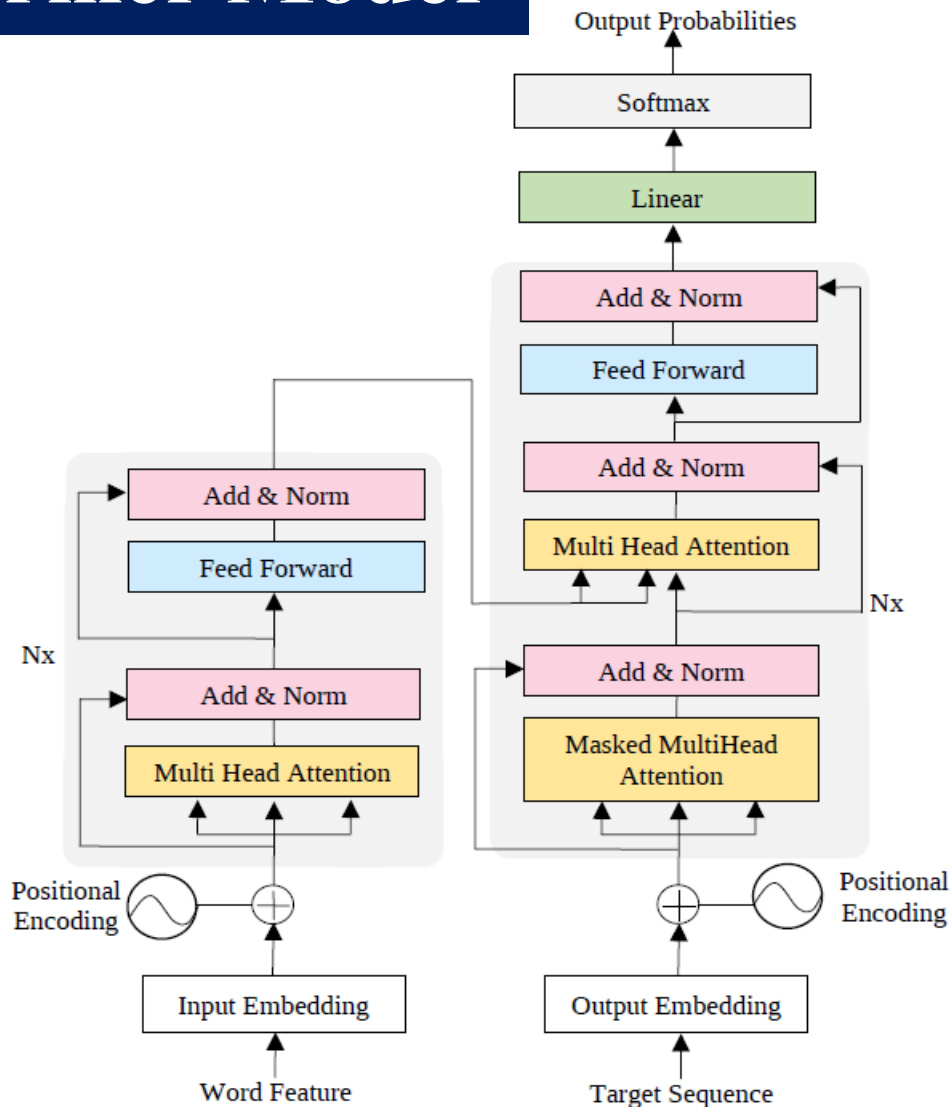
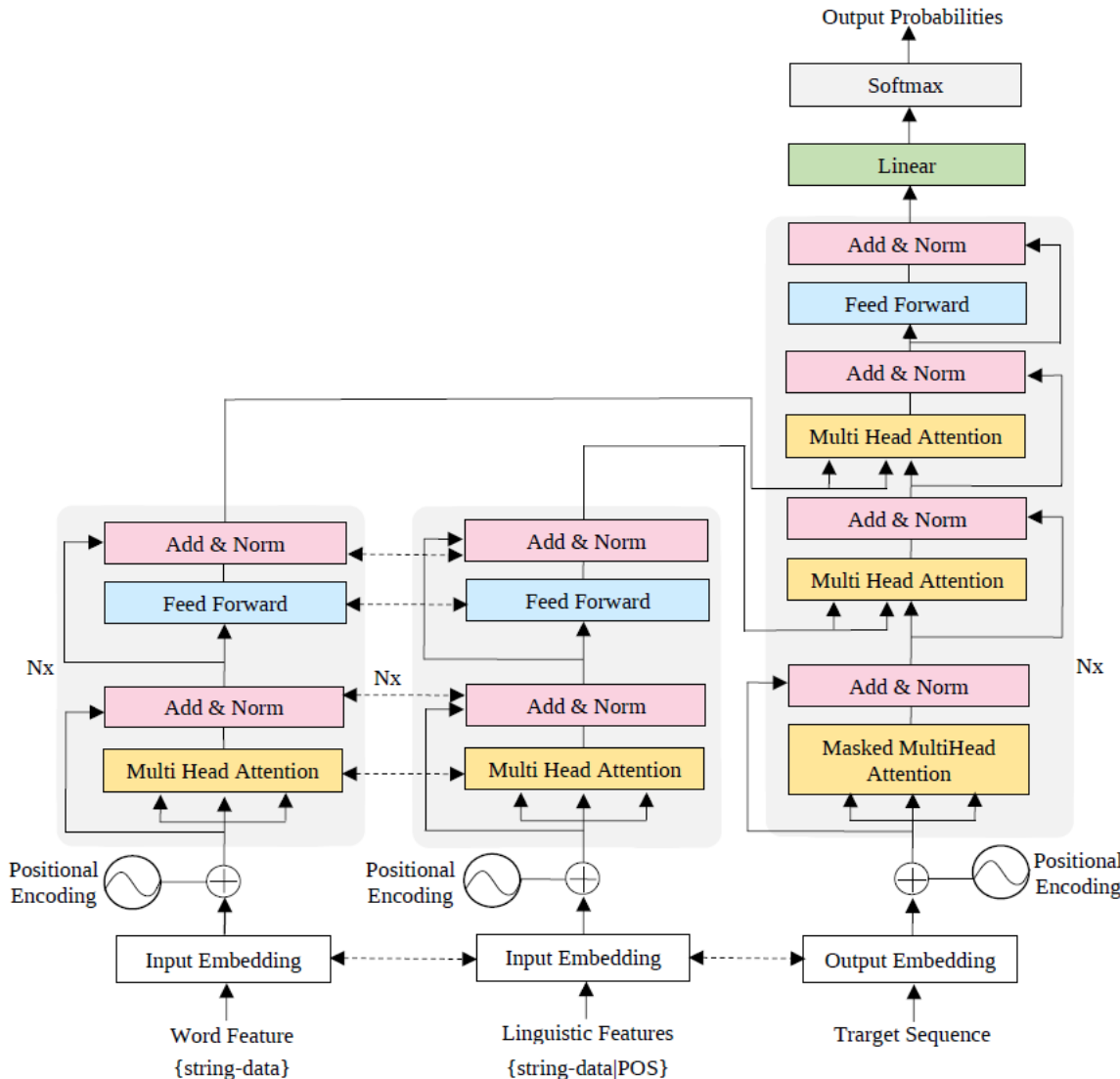


Figure 4.1 Transformer Architecture (Vaswani et al., 2017)

Multi-Source Transformer Model



- composed of two encoders and a decoder
- dash arrows indicate sharing parameters
- produce different activations and combine
- two multi-head attention components, one for each encoder

- decoder produces target sequence
 - $\{string-data\}$
 - $\{string-data|POS\}$
- multi-source model is fed with a tri-parallel corpus
 - input $\{string-data\}$
 - input $\{string-data|POS\}$
 - target $\{string-data\}$ or $\{string-data|POS\}$
- both inputs are processed simultaneously during translation

Figure 4.2 Architecture of Multi-source Transformer

The background features a dark blue overlay on a photograph of a large, multi-story university building with a prominent portico and columns. To the right, there is a large, stylized white hexagonal graphic element that partially overlaps the building image. A smaller, solid dark blue hexagon is positioned above it.

Preprocessing

Building Myanmar POS tagger

- Need to build Myanmar POS tagger
- There is only one available small POS-tagged corpus (myPOS corpus version 1.0)
- Corpus is small and doesn't cover some domain areas
- Extended myPOS version 1.0 to myPOS version 2.0 and built POS tagging models
- Compared the tagging accuracies between four models
 - i. Conditional Random Fields (CRFs)
 - ii. Hidden Markov Model (HMM)
 - iii. Ripple Down Rules-based (RDR)
 - iv. Neural Sequence labeling Model (NCRF++)

POS Tag Sets

Table 5.1 Part-of-Speech Tag Sets for Myanmar Language

No.	POS Tag	Brief Definition
1.	abb	Abbreviation
2.	adj	Adjective
3.	adv	Adverb
4.	conj	Conjunction
5.	fw	Foreign Word
6.	int	Interjection
7.	n	Noun
8.	num	Number
9.	part	Particle
10.	ppm	Post-positional Marker
11.	pron	Pronoun
12.	punc	Punctuation
13.	sb	Symbol
14.	tn	Text Number
15.	v	Verb

Conditional Random Fields (CRFs)

- Build a probabilistic model for segmenting and labeling sequence data

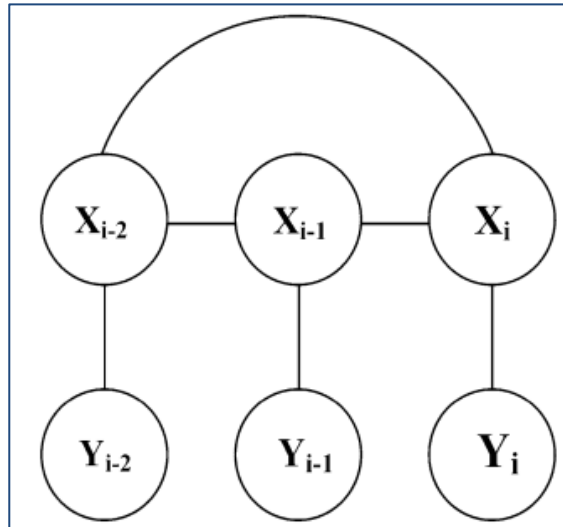


Figure 5.1 Graphical representation of CRF model

- Main objective is to identify probable tag or label for word sequence based on conditional probability
- Conditional probability is defined as:

$$P(Y|X) = \frac{1}{Z_X} \exp\left[\sum_i \sum_j \lambda_j f_j(y_{i-1}, y_i, X, i)\right]$$

Hidden Markov Model (HMM)

- Computes probability distribution over possible sequences of labels
- Markov chains are the basis for HMMs
- Model that specifies a sequence of potential events
- Markov model is based on Markov assumption

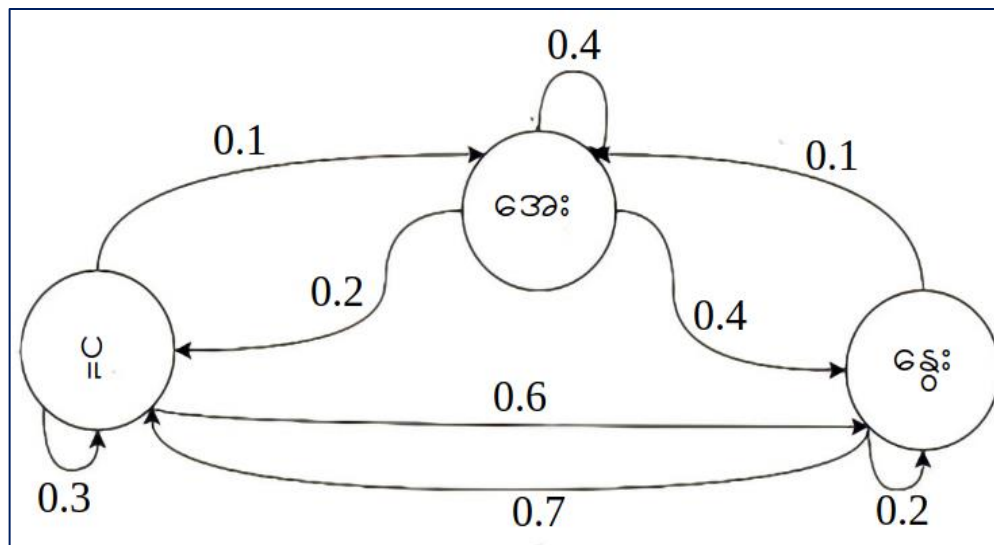


Figure 5.2 A Markov chain with states and transitions

Hidden Markov Model (HMM)

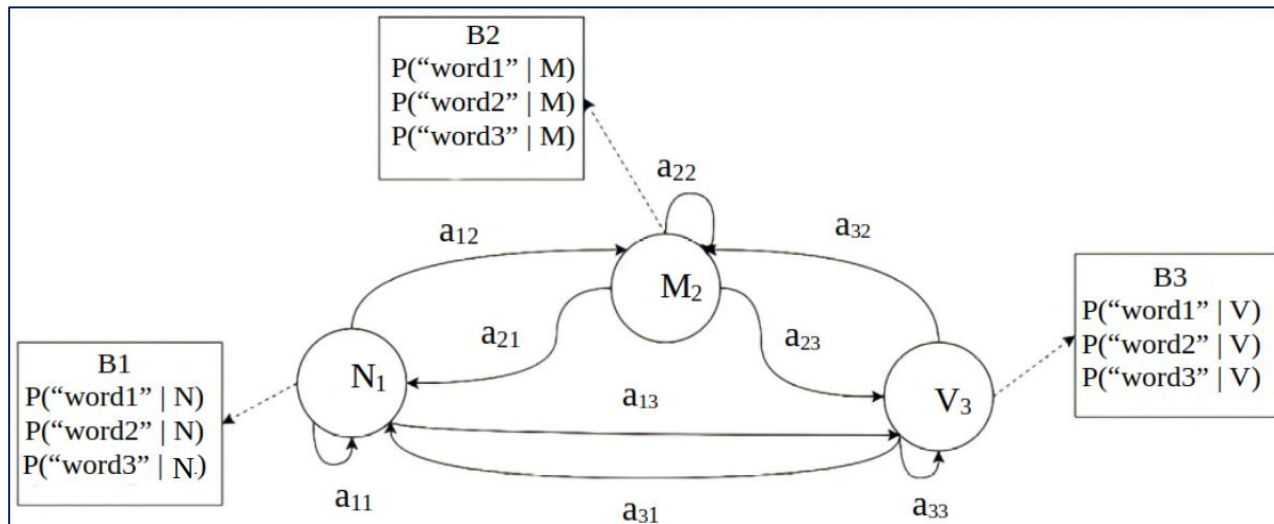


Figure 5.3 An HMM with A transition and B emission probabilities

- Attempts to decode the correct POS tag for the three words from state
- A transition probabilities move from one state to another
- B emission probabilities indicate how possible

Ripple Down Rules-based (RDR)

- An approach to build knowledge-based systems
- RDRPOSTagger provides a failure-driven approach
- SCRDR trees are binary trees with two distinct edges:
 - i. except edge*
 - ii. if-not edge*
- Every node in a tree has a *rule* associated with it
- Rule is expressed as: if X then Y where X is condition and Y is conclusion

Ripple Down Rules-based (RDR)

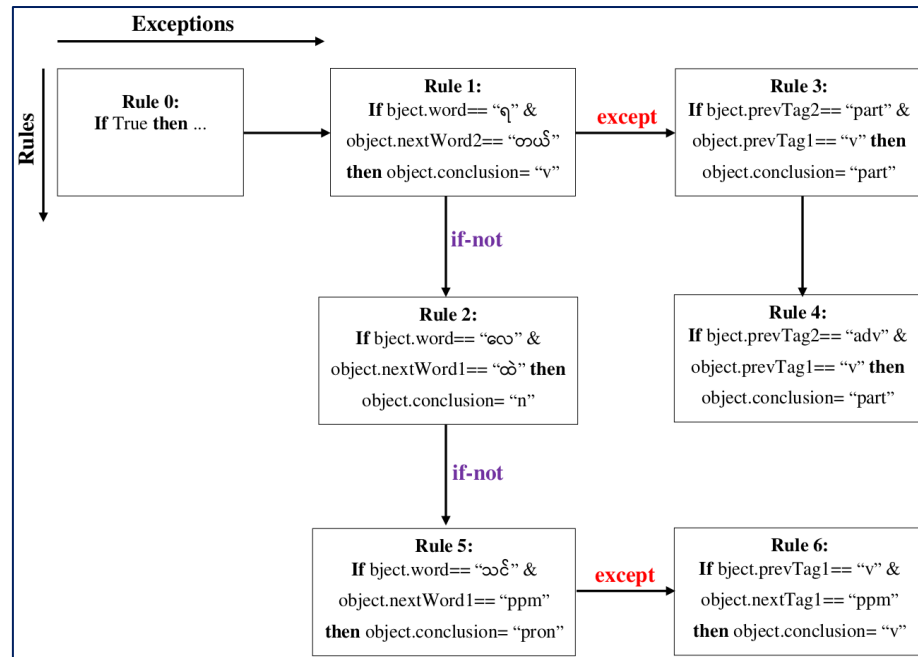


Figure 5.4 A binary tree of single classification ripple down rules

- Shows SCRDR tree for a case “ကျွန်တော်/pron သင်ခန်းစာ/n တစ်/t၏ ခု/part ကို/part လေ့လာ/v ရ/v တယ်/ppm ။/punc” (“I learned a lesson.” in English)
- POS tag for the word “ရ” as “part”

Neural Sequence Labeling Model (NCRF++)

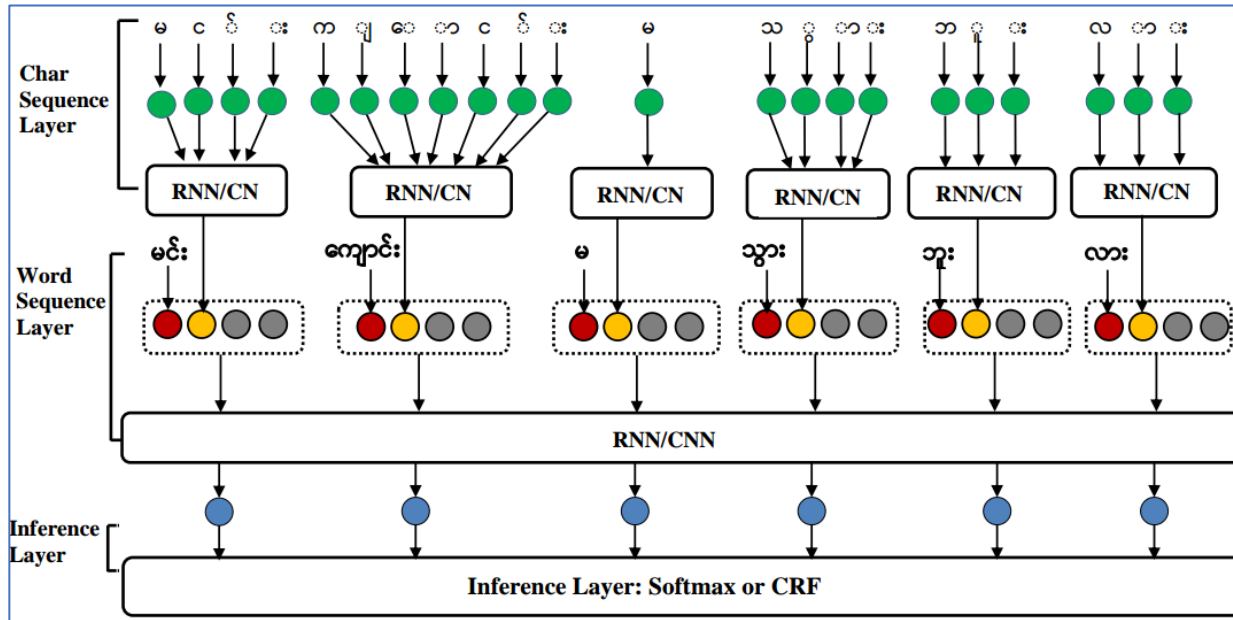


Figure 5.5 NCRF ++ for Burmese sentence “မင်းကျောင်းမသွားဘူးလား” (Don't you go to school?).

- words are represented with word embeddings
- character sequence layer can be used to automatically extract word-level features
- concatenation of word embeddings, character sequence representations, and handcrafted features
- takes word representations as input and extracts sentence level features

Toolkits used in POS Tagging Experiment

- i. *CRFsuite* tool version 0.12
(Okazaki,2007) <https://github.com/chokkan/crfsuite>
- ii. NCRF++ toolkit for neural sequence labeling (Yang et al., 2018) <https://github.com/jiesutd/NCRFpp>
- iii. Jitar (version 0.3.3) based on a trigram Hidden Markov Model (HMM)
- iv. RDRPOSTagger (Version 1.2.3) Part-of-Speech and morphological tagging toolkit (D. Q. Nguyen et al., 2014)

Corpus Extension

- extended original myPOS corpus from Myanmar-Chinese parallel corpus and Myanmar-Korean parallel corpus as myPOS corpus version 2.0
- word segmentation, POS tagging for each word and error checking was done manually
- firstly, checked and corrected manually the segmented Myanmar sentences from the above parallel corpora
- two-segmented corpora are manually tagged and checked repeatedly

Corpus Extension

- we defined 10K POS tagged Myanmar sentences of [Myanmar-Chinese Parallel Corpus](#) as [Extension-1](#)
- another 11K POS tagged Myanmar sentences of [Myanmar-Korean Parallel Corpus](#) as [Extension-2](#)
- these two different domains of POS tagged sentences are combined with original myPOS corpus to obtain the extended [myPOS corpus version 2.0](#)
- the extended corpus becomes 31,052 sentences, which approximately triple size of the original myPOS corpus

Corpus Extension

Table 5.2 Data Comparison between Original Baseline Corpus and Extended Corpus (myPOS corpus version 2.0)

Count	Original Corpus	Extended Corpus		
	myPOS	Extension-1	Extension-2	myPOS version-2
Total no. of Sentences	11,000	10,000	10,052	31,052
Total no. of Words	239,598	103,909	106,864	450,371
Average Words per Sentence	21.78	10.39	10.63	14.50

Evaluation for POS Tagging Model

- Measured the POS tagging performance by using the *Accuracy* defined in the following equation:

$$\text{Accuracy} = \frac{\text{number of correct POS tags}}{\text{number of tokens in test corpus}}$$

Tagging Accuracy

Table 5.3 Comparison of accuracies between the original myPOS corpus and the extended myPOS corpus version 2.0

Methods	cest: Closed Test-set		otest-1		otest-2	
	Original myPOS	myPOS (Version 2.0)	Original myPOS	myPOS (Version 2.0)	Original myPOS	myPOS (Version 2.0)
CRFs	98.40%	98.19%	91.69%	94.61%	91.79%	94.75%
NCRF++ (CharLSTM+WordCNN+CRF)	96.50%	95.10%	91.07%	91.65%	90.15%	94.64%
NCRF++ (CharCNN+WordLSTM)	97.61%	97.40%	93.14%	95.79%	93.23%	95.98%
HMM	97.14%	96.44%	93.66%	94.97%	93.78%	95.06%
RDR	98.43%	98.35%	95.10%	97.54%	95.15%	97.57%

Building New Thai POS Tagger

- LST20 POS tagged corpus (Prachya et al., 2020) were used to build a new Thai POS tagger
- The corpus consisted of 74,180 sentences, and was annotated with 16 distinct POS tags

Table 5.4 Part-of-Speech Tag Sets for Thai Language

No.	POS Tag	Brief Definition
1.	AJ	Adjective
2.	AV	Adverb
3.	AX	Auxiliary
4.	CC	Connector
5.	CL	Classifier
6.	FX	Prefix
7.	IJ	Interjection
8.	NG	Negation
9.	NN	Noun
10.	NU	Number
11.	PA	Particle
12.	PR	Pronoun
13.	PS	Preposition
14.	PU	Punctuation
15.	VV	Verb
16.	XX	Others

Building New Thai POS Tagger

- Existing RDRPOSTagger used 44 POS tags for Thai POS tagger and 45 POS tags for English POS tagger
- Due to different numbers of POS tags, the translation performance may differ according to each tagger
- LST20 dataset format is CoNLL2003 style format with four columns (Word, POS tag, Named entity, and Clause boundary)
- Word and POS tag columns were retrieved and converted into sentences

Building New Thai POS Tagger

- We used the converted POS tagged sentences (over 74K) as training data and 1,000 sentences for test data
- RDRPOSTagger toolkit (*D. Q. Nguyen et al.*, 2014) was used for building Thai POS tagger
- Our newly built Thai POS tagger provides a tagging accuracy of 96.61%

The background features a dark, blue-tinted photograph of a large, multi-story university building with a prominent portico and columns. Overlaid on the right side of the image are several dark blue geometric shapes: a large, irregular polygon and two hexagons. A white rectangular box with a thin border is positioned in the center-left, containing the text.

Experiments and Results

Parallel Data

- Thai, English, and Myanmar sentences from ASEAN-MT Corpus (Prachya & Thepchai, 2013)
- Corpus is a parallel corpus in travel domain and it is composed of six main categories:
 - people* (greeting, introduction, and communication),
 - survival* (transportation, accommodation, and finance),
 - food* (food, beverage, and restaurant),
 - fun* (recreation, traveling, shopping, and nightlife),
 - resources* (number, time, and accuracy), and
 - special needs* (emergency and health)

Table 6.1 Parallel dataset for Thai, English, and Myanmar

Data	File name	No. of sentences	No. of words	No. of unique words
TH	train.th	20,000	139,767	8,710
	valid.th	1,013	6,809	1,376
	test.th	1,000	7,169	1,363
EN	train.en	20,000	141,098	12,553
	valid.en	1,013	7,245	2,113
	test.en	1,000	7,176	2,201
MY	train.my	20,000	187,547	7,712
	valid.my	1,013	9,573	1,428
	test.my	1,000	9,579	1,439

Data Preprocessing

- Preprocessing steps are required before conducting the training process
- Although Thai and English sentences are correctly segmented in ASEAN-MT corpus, Myanmar sentences are not
- We used *myWord* segmenter for Myanmar word segmentation
- POS tagging was performed on each dataset to apply NMT models that utilized linguistic features

Part-of-Speech Tagging

- We used our built RDR POS tagger for Myanmar tagged data
- English and Thai data were tagged using RDRPOSTagger toolkit (*D. Q. Nguyen et al.*, 2014)
- We also used our newly built Thai POS tagger to compare translation models that utilized existing Thai POS tagger
- In our experiment, we used both existing and our newly built Thai POS taggers

Universal Part-of-Speech (UPOS) Tags

Table 6.2 Universal Part-of-Speech Tag Sets

No.	POS Tag	Brief Definition
1.	NOUN	nouns
2.	VERB	verbs
3.	ADJ	adjective
4.	ADV	adverbs
5.	PRON	pronouns
6.	DET	determiners and articles
7.	ADP	prepositions and postpositions
8.	NUM	numerals
9.	CONJ	conjunctions
10.	PRT	particles
11.	".."	punctuation marks
12.	X	abbreviations or foreign words

- UPOS tags are part-of-speech tags used in UD framework for consistent annotation of grammar
- POS and UPOS tags are incorporated into NMT models
- We used 12 UPOS tags defined by Petrov et al. (2012)

Universal Part-of-Speech (UPOS) Tags

For UPOS tagged data, we used our built *POS-to-UPOS* tags converter

Table 6.3 Examples of POS and UPOS tagged sentences

Thai Sentence	ฉัน	ต้องการ	ซอง	จดหมาย			
Original POS tags by <i>existing tagger</i>	PPRS	VACT	NCMN	NCMN			
Universal POS tags	PRON	VERB	NOUN	NOUN			
Original POS tags by <i>new tagger</i>	PR	VV	NN	NN			
Universal POS tags	PRON	VERB	NOUN	NOUN			
English Sentence	I	need	an	envelope.			
Original POS tags	PRP	VBP	DT	NN			
Universal POS tags	PRON	VERB	DET	NOUN			
Myanmar Sentence	ကျွန်တော်	စာအိတ်	တစ်	အိတ်	လို့	တယ်	။
Original POS tags	<u>pron</u>	n	<u>tn</u>	n	v	ppm	<u>punc</u>
Universal POS tags	PRON	NOUN	NUM	NOUN	VERB	ADP	.

Model Configuration and Evaluation

- Conducted several experiments on two-way translation of TH–EN, TH–MY, and EN–MY pairs
- Used *Marian* framework (Junczys-Dowmunt et al., 2018)
- Experiments are carried out using *transformer*, *multi-source transformer*, and *shared-multi-source transformer* models
- For proposed models, we integrated linguistic features into each word on source or target side or both

Model Configuration and Evaluation

- used *transformer* model that utilized only word vector as first baseline model
- Edit-Based Transformer with Repositioning: EDITOR (*Xu & Carpuat, 2021*) as second baseline model
 - non-autoregressive transformer model that iteratively edits hypothesis using novel reposition operation
- can implement EDITOR with soft or hard lexical constraints or without constraints
- the authors stated that EDITOR perform better with soft constraints
- experimented EDITOR model with soft constraints by Pytorch-based *FairSeq* framework

Model Configuration and Evaluation

- test data are translated and statistical significance evaluations are computed using *compare-mt* tool (*Neubig et al.*, 2019)
- evaluation results are described in terms of *case-insensitive Bilingual Evaluation Understudy* (BLEU) and *character n-gram F-score* (chrF)

Model Configuration and Evaluation

BLEU

- *de facto* standard automatic evaluation metric
- much faster and cheaper than human measure
- BLEU algorithm is language independent
- measures precision of *n-gram* (4-gram in our case) with respect to reference translation
- BLEU is calculated as:

$$BLEU = \min\left(1, \frac{\text{output-length}}{\text{reference-length}}\right) \left(\prod_{i=1}^n \text{precision}_i\right)^{\frac{1}{n}}$$

Model Configuration and Evaluation

chrF

- character n -gram F-score for automatic evaluation of MT output
- performs better than the *wordF* score (M. Popović, 2016)
- *main advantage*: does not penalize too different variants of acceptable translations
- *chrF* score is computed as:

$$\text{chrF} = (1 + \beta^2) \frac{\text{chrP} \cdot \text{chrR}}{\beta^2 \cdot \text{chrP} + \text{chrR}}$$

Translation Results

- Used short notations for our proposed models
- Factored models are denoted by the following notations:
 - “t” stands for “translation,”
 - “W” stands for “words,”
 - “P” stands for “part-of-speech (POS) tags,” and
 - “UP” stands for “universal part-of-speech (UPOS) tags”

Table 6.4 Experimented Models for each language pair

No.	Models
1.	Baseline Transformer - $t(W-W)$
2.	Baseline EDITOR with soft constraints
3.	Transformer - $t(W P\text{-to-}W)$
4.	Transformer - $t(W\text{-to-}W P)$
5.	Transformer - $t(W P\text{-to-}W P)$
6.	Multi-Source Transformer - $t(W+W P\text{-to-}W)$
7.	Multi-Source Transformer - $t(W+W P\text{-to-}W P)$
8.	Shared-Multi-Source Transformer - $t(W+W P\text{-to-}W)$
9.	Shared-Multi-Source Transformer - $t(W+W P\text{-to-}W P)$
10.	Transformer - $t(W UP\text{-to-}W)$
11.	Transformer - $t(W\text{-to-}W UP)$
12.	Transformer - $t(W UP\text{-to-}W UP)$
13.	Multi-Source Transformer - $t(W+W UP\text{-to-}W)$
14.	Multi-Source Transformer - $t(W+W UP\text{-to-}W UP)$
15.	Shared-Multi-Source Transformer - $t(W+W UP\text{-to-}W)$
16.	Shared-Multi-Source Transformer - $t(W+W UP\text{-to-}W UP)$

Translation Results

Table 6.5 Experimental results for Thai-to-Myanmar translation pair.

Models	Thai-to-Myanmar by Existing Thai POS Tagger (44 POS tags)		Thai-to-Myanmar by New Thai POS Tagger (16 POS tags)	
	BLEU scores (%)	chrF scores (%)	BLEU scores (%)	chrF scores (%)
Baseline Transformer - t(W-W)	22.90	39.75	22.90	39.75
Baseline EDITOR with soft constraints	18.14	41.80	18.14	41.80
Multi-Source Transformer - t(W+W P-to-W)	24.81[†]	40.62	24.75[†]	40.48
Multi-Source Transformer - t(W+W P-to-W P)	25.34[†]	41.43 [*]	25.03[†]	41.42 [*]
Shared-Multi-Source Transformer - t(W+W P-to-W)	24.17[†]	40.68	25.36[†]	42.17[*]
Shared-Multi-Source Transformer - t(W+W P-to-W P)	24.91[†]	41.43 [*]	24.37[†]	41.82[*]
Multi-Source Transformer - t(W+W UP-to-W)	24.64[†]	40.79	24.36[†]	40.62
Multi-Source Transformer - t(W+W UP-to-W UP)	24.02[†]	39.76	24.89[†]	41.00 [*]
Shared-Multi-Source Transformer - t(W+W UP-to-W)	24.60[†]	41.77 [*]	25.01[†]	41.35 [*]
Shared-Multi-Source Transformer - t(W+W UP-to-W UP)	25.12[†]	41.82[*]	25.25[†]	42.29[*]

Bold numbers indicate higher BLEU and chrF scores than the two baseline models. Scores with an asterisk (*) are significantly higher than the first baseline Transformer model and scores with a dagger (†) are significantly higher than the second baseline EDITOR model at $p < 0.05$.

Translation Results

Table 6.6 Experimental results for Myanmar-to-Thai translation pair

Models	Myanmar-to-Thai by Existing Thai POS Tagger (44 POS tags)		Myanmar-to-Thai by New Thai POS Tagger (16 POS tags)	
	BLEU scores (%)	chrF scores (%)	BLEU scores (%)	chrF scores (%)
Baseline Transformer - t(W-W)	24.92	41.73	24.92	41.73
Baseline EDITOR with soft constraints	14.30	42.94	14.30	42.94
Transformer - t(W-to-W P)	25.18[†]	41.37	24.44 [†]	40.84
Multi-Source Transformer - t(W+W P-to-W)	24.77 [†]	43.79*	24.77 [†]	43.79*
Multi-Source Transformer - t(W+W P-to-W P)	24.02 [†]	43.49*	24.56 [†]	43.32*
Shared-Multi-Source Transformer - t(W+W P-to-W)	24.79 [†]	44.41*	24.79 [†]	44.41*
Multi-Source Transformer - t(W+W UP-to-W)	24.92 [†]	43.25*	24.92 [†]	43.25*
Multi-Source Transformer - t(W+W UP-to-W UP)	24.71 [†]	43.11	24.51 [†]	43.16
Shared-Multi-Source Transformer - t(W+W UP-to-W UP)	24.27 [†]	43.13	25.02[†]	43.85*

Bold numbers indicate higher BLEU and chrF scores than the two baseline models. Scores with an asterisk (*) are significantly higher than the first baseline Transformer model and scores with a dagger (†) are significantly higher than the second baseline EDITOR model at $p < 0.05$.

Translation Results

Table 6.7 Experimental results for Myanmar-to-English and English-to-Myanmar translation pairs

Models	Myanmar-to-English		English-to-Myanmar	
	BLEU scores (%)	chrF scores (%)	BLEU scores (%)	chrF scores (%)
Baseline Transformer - t(W-W)	27.76	44.30	27.16	44.12
Baseline EDITOR with soft constraints	17.14	43.15	24.33	46.71
Multi-Source Transformer - t(W+W P-to-W)	28.76[†]	46.61^{*†}	31.72^{*†}	48.03[*]
Multi-Source Transformer - t(W+W P-to-W P)	28.86[†]	46.64^{*†}	31.53^{*†}	48.03[*]
Shared-Multi-Source Transformer - t(W+W P-to-W)	28.93[†]	47.08^{*†}	31.24^{*†}	47.62[*]
Shared-Multi-Source Transformer - t(W+W P-to-W P)	29.01[†]	46.81^{*†}	31.26^{*†}	47.66[*]
Multi-Source Transformer - t(W+W UP-to-W)	28.86[†]	46.72^{*†}	31.66^{*†}	48.30[*]
Multi-Source Transformer - t(W+W UP-to-W UP)	28.39[†]	45.93^{*†}	31.67^{*†}	47.98[*]
Shared-Multi-Source Transformer - t(W+W UP-to-W)	29.84^{*†}	48.08^{*†}	31.73^{*†}	48.53[*]
Shared-Multi-Source Transformer - t(W+W UP-to-W UP)	29.31[†]	47.08^{*†}	31.26^{*†}	47.74[*]

Bold numbers indicate higher BLEU and chrF scores than the two baseline models. Scores with an asterisk (*) are significantly higher than the first baseline Transformer model and scores with a dagger (†) are significantly higher than the second baseline EDITOR model at $p < 0.05$.

Translation Results

Table 6.8 Experimental results for Thai-to-English translation pair

Models	Thai-to-English by Existing Thai POS Tagger (44 POS tags)		Thai-to-English by New Thai POS Tagger (16 POS tags)	
	BLEU scores (%)	chrF scores (%)	BLEU scores (%)	chrF scores (%)
Baseline Transformer - t(W-W)	30.94	45.64	30.94	45.64
Baseline EDITOR with soft constraints	23.78	46.47	23.78	46.47
Multi-Source Transformer - t(W+W P-to-W)	34.89^{*†}	49.56^{*†}	34.83^{*†}	49.33^{*†}
Multi-Source Transformer - t(W+W P-to-W P)	33.80^{*†}	48.58^{*†}	34.26^{*†}	49.78^{*†}
Shared-Multi-Source Transformer - t(W+W P-to-W)	34.22^{*†}	49.70^{*†}	35.51^{*†}	50.69^{*†}
Shared-Multi-Source Transformer - t(W+W P-to-W P)	34.25^{*†}	49.81^{*†}	34.71^{*†}	49.57^{*†}
Multi-Source Transformer - t(W+W UP-to-W)	34.98^{*†}	49.84^{*†}	34.63^{*†}	49.39^{*†}
Multi-Source Transformer - t(W+W UP-to-W UP)	34.39^{*†}	49.62^{*†}	34.94^{*†}	50.17^{*†}
Shared-Multi-Source Transformer - t(W+W UP-to-W)	34.78^{*†}	50.05^{*†}	35.28^{*†}	50.51^{*†}
Shared-Multi-Source Transformer - t(W+W UP-to-W UP)	35.63^{*†}	49.96^{*†}	34.55^{*†}	50.40^{*†}

Scores with an asterisk (*) are significantly higher than the first baseline Transformer model and scores with a dagger (†) are significantly higher than the second baseline EDITOR model at $p < 0.05$.

Translation Results

Table 6.9 Experimental results for English-to-Thai translation pair

Models	English-to-Thai by Existing Thai POS Tagger (44 POS tags)		English-to-Thai by New Thai POS Tagger (16 POS tags)	
	BLEU scores (%)	chrF scores (%)	BLEU scores (%)	chrF scores (%)
Baseline Transformer - t(W-W)	31.04	48.81	31.04	48.81
Baseline EDITOR with soft constraints	24.60	51.28	24.60	51.28
Transformer - t(W-to-W P)	30.92 [†]	48.49	30.16 [†]	47.89
Multi-Source Transformer - t(W+W P-to-W)	35.50^{*†}	53.67^{*†}	35.50^{*†}	53.67^{*†}
Multi-Source Transformer - t(W+W P-to-W P)	35.67^{*†}	53.66^{*†}	34.76^{*†}	53.15^{*†}
Shared-Multi-Source Transformer - t(W+W P-to-W)	36.73^{*†}	54.40^{*†}	36.73^{*†}	54.40^{*†}
Shared-Multi-Source Transformer - t(W+W P-to-W P)	35.19^{*†}	53.16^{*†}	34.98^{*†}	53.31^{*†}
Multi-Source Transformer - t(W+W UP-to-W)	36.40^{*†}	53.92^{*†}	36.40^{*†}	53.92^{*†}
Multi-Source Transformer - t(W+W UP-to-W UP)	34.67^{*†}	53.62^{*†}	34.41^{*†}	53.13^{*†}
Shared-Multi-Source Transformer - t(W+W UP-to-W)	37.03^{*†}	54.06^{*†}	37.03^{*†}	54.06^{*†}
Shared-Multi-Source Transformer - t(W+W UP-to-W UP)	35.50^{*†}	53.45^{*†}	34.85^{*†}	53.47^{*†}

- transformer models that incorporate linguistic features improved translation performance
- All experimental results answer our research questions regarding whether adding linguistic features to transformer-based models improves the translation quality for two-way translation of low-resource translation pairs

Translation Errors and Analysis

Table 6.10 Samples translation for the proposed models and the two baselines on Thai-to-Myanmar translation.

Thai-to-Myanmar Translation	
Source	เมื่อไหร่ มัน จะ ถึง พิพิธภัณฑ ?
Reference	ပြု တိုက် ကို ဘယ် အ ချိန် ရောက် မှာ လဲ ။
Baseline Transformer - $t(W-W)$	ဘယ် အ ချိန် လောက် ရောက် ရ မ လဲ ။
Baseline EDITOR with soft constraints	ပြု တိုက် ရောက် မ လဲ ။
Transformer - $t(W P\text{-to-}W P)$	ပြု တိုက် ကို ဘယ် အ ချိန် ရောက် မှာ လဲ ။
Multi-Source Transformer - $t(W+W P\text{-to-}W P)$	ပြု တိုက် ကို ဘယ် အ ချိန် ရောက် မှာ လဲ ။
Shared-Multi-Source Transformer - $t(W+W P\text{-to-}W)$	ပြု တိုက် ကို ဘယ် အ ချိန် ရောက် မှာ လဲ ။

Word Error Analysis

- WER was used to analyze translated outputs
- Thai-to-English and English-to-Thai translation pairs were selected to compare WER scores between the models
- WER can be calculated as:

$$\text{WER} = \frac{(I + D + S) \times 100}{N}$$

Word Error Analysis

Table 6.11 Comparison of WER scores for Thai-to-English and English-to-Thai translation pairs.

Models	Thai-to-English		English-to-Thai	
	WER (%) by Existing Thai POS Tagger	WER (%) by New Thai POS Tagger	WER (%) by Existing Thai POS Tagger	WER (%) by New Thai POS Tagger
Baseline Transformer - t(W-W)	54.80	54.80	62.40	62.40
Baseline EDITOR with soft constraints	54.40	54.40	60.60	60.60
Multi-Source Transformer - t(W+W P-to-W)	49.50	<u>49.30</u>	55.90	55.90
Multi-Source Transformer - t(W+W P-to-W P)	50.70	51.20	55.30	56.30
Shared-Multi-Source Transformer - t(W+W P-to-W)	50.60	<u>49.30</u>	<u>53.80</u>	<u>53.80</u>
Shared-Multi-Source Transformer - t(W+W P-to-W P)	50.80	49.50	56.40	55.90
Multi-Source Transformer - t(W+W UP-to-W)	<u>49.00</u>	50.60	55.10	55.10
Multi-Source Transformer - t(W+W UP-to-W UP)	50.50	50.60	54.90	55.40
Shared-Multi-Source Transformer - t(W+W UP-to-W)	50.00	50.00	54.10	54.10
Shared-Multi-Source Transformer - t(W+W UP-to-W UP)	49.10	50.70	55.60	56.00

Word Error Analysis

Table 6.12 Top 10 confusion pairs of Thai-to-English translation.

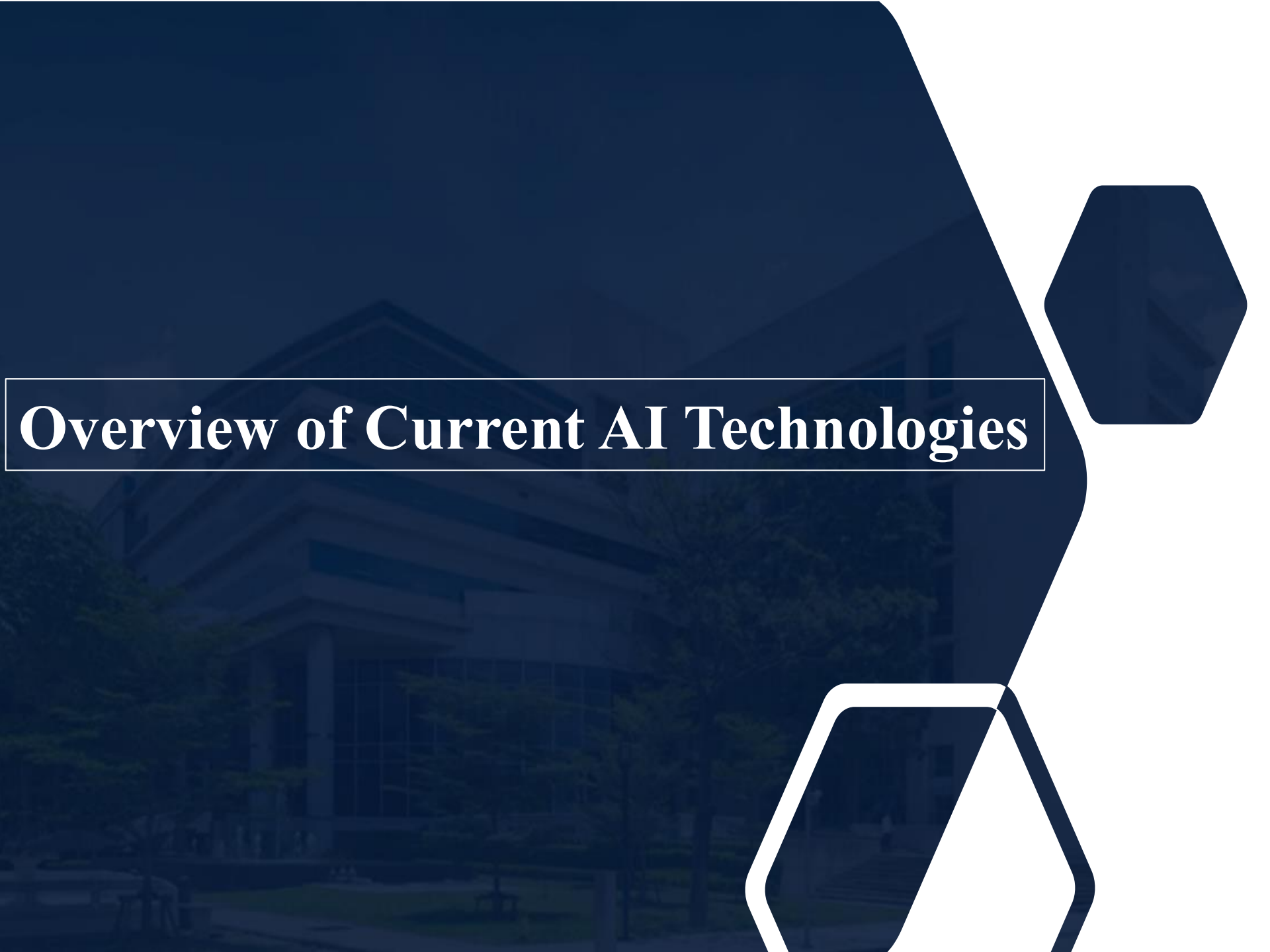
Baseline Transformer - t(W-W)		Baseline EDITOR with soft constraints		Shared-Multi-Source Transformer - t(W+W P-to-W P)	
Frequency	Ref → Hyp	Frequency	Ref → Hyp	Frequency	Ref → Hyp
11	the → a	8	a → the	8	are → is
6	like → want	4	does → is	7	like → want
8	are → is	4	the → a	6	want → like
5	for → to	4	ticket → tickets	5	can → may
5	want → like	3	for → to	5	the → a
4	can → may	3	do → would	4	a → an
3	a → an	3	like → want	4	can → could
3	an → a	3	luggage → suitcase	4	have → do
3	can → could	3	are → is	3	a → the
3	could → can	3	in → to	2	certainly → sure



Summary of Research Work

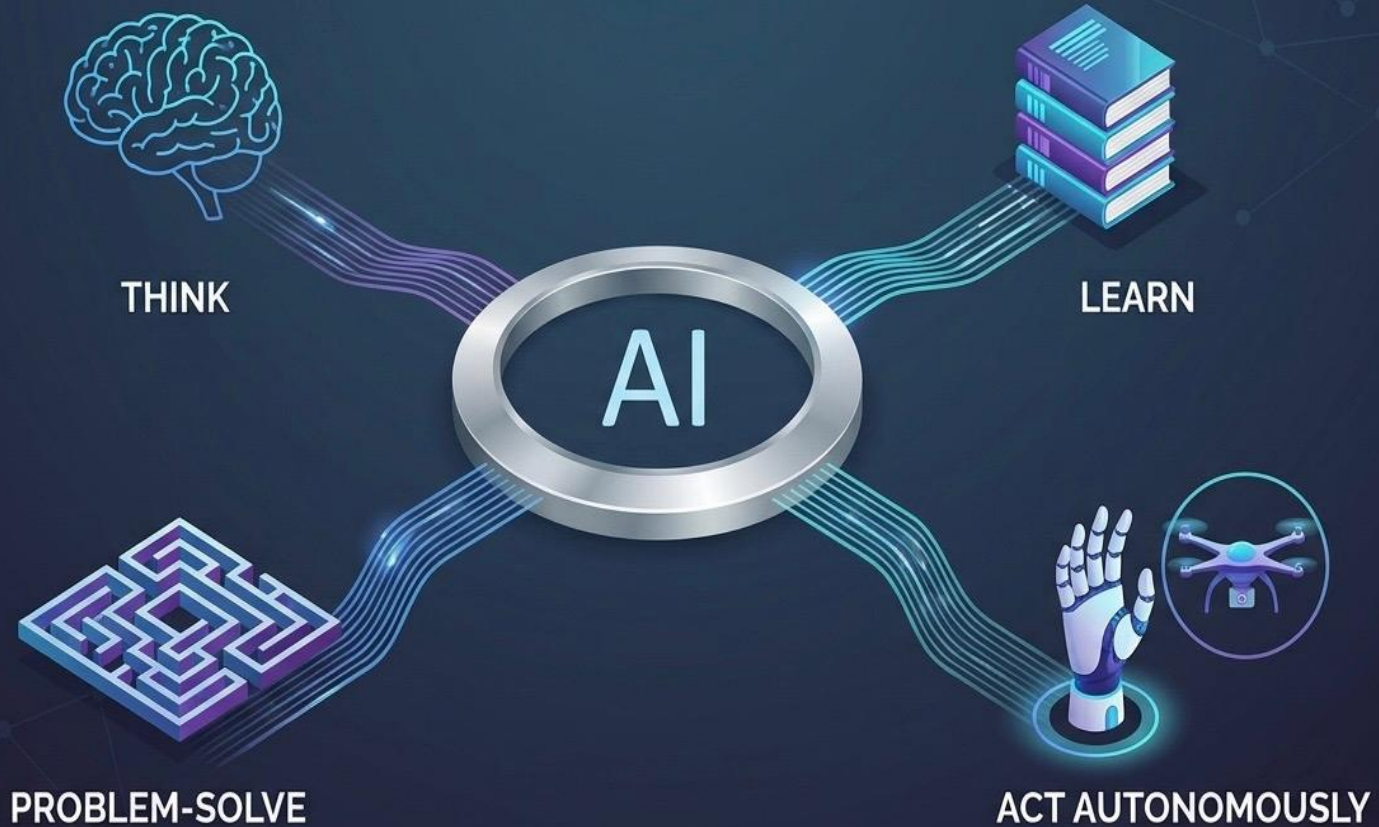
Summary

- We proposed NMT systems that incorporate linguistic features for the translation of Thai-to-Myanmar, Thai-to-English, Myanmar-to-English and vice versa
- The transformer, multi-source transformer, and shared-multi-source transformer models were used
- Experimental results show that integrating linguistic features in NMT models helps to improve the translation performance for all language pairs
- Our proposed models achieved better translation quality compared with the baseline transformer and EDITOR models
- *Shared-multi-source* transformer model provided the highest significant translation quality based on the scores.

The background features a dark, blue-tinted photograph of a large, modern building with a prominent portico and columns. Overlaid on the right side of the image is a large, dark blue hexagonal graphic element. A white hexagonal outline is positioned at the bottom right, partially overlapping the dark blue hexagon.

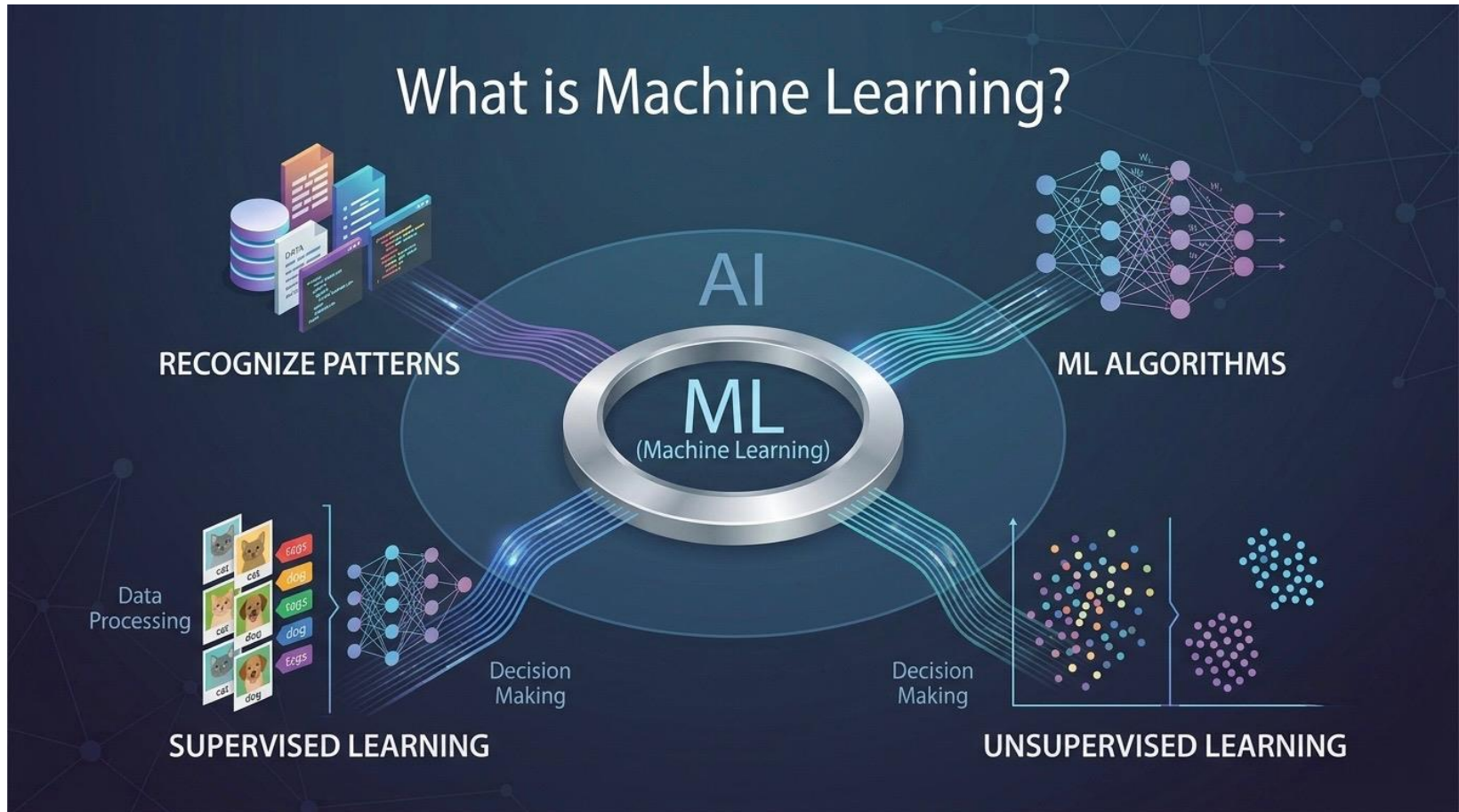
Overview of Current AI Technologies

What is AI?



- Artificial Intelligence (AI) is the field of computer science that aims to create systems capable of performing tasks that normally require human intelligence.

What is Machine Learning?

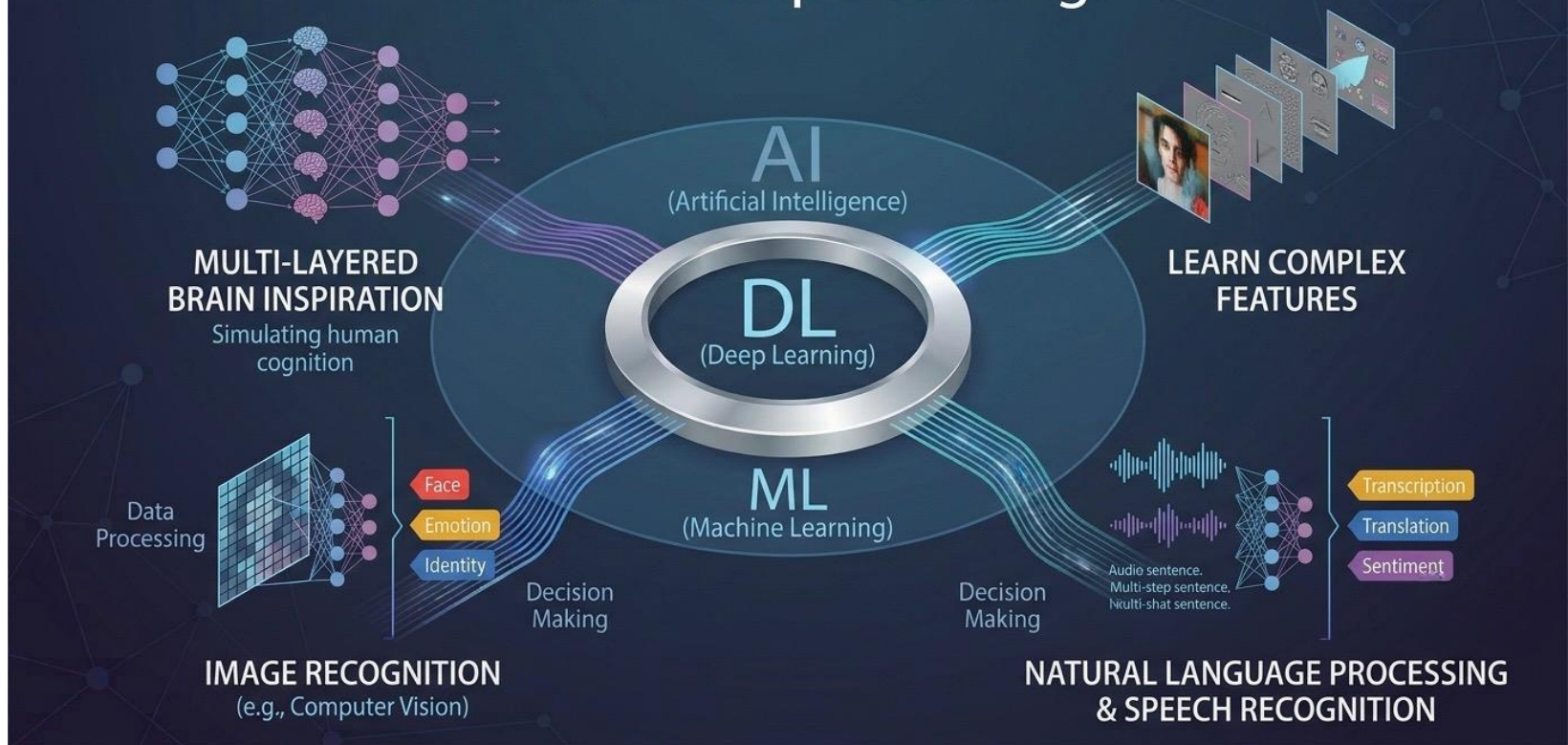


- **Machine Learning (ML)** is a **subset of Artificial Intelligence (AI)** that enables machines to **learn from data** without being explicitly programmed.
- Instead of following fixed rules, ML algorithms identify **patterns** in data and use them to make **predictions or decisions**.

Types of Machine Learning

1. Supervised Learning
 - Learns from **labeled data**
2. Unsupervised Learning
 - Learns from **unlabeled data**

What is Deep Learning?

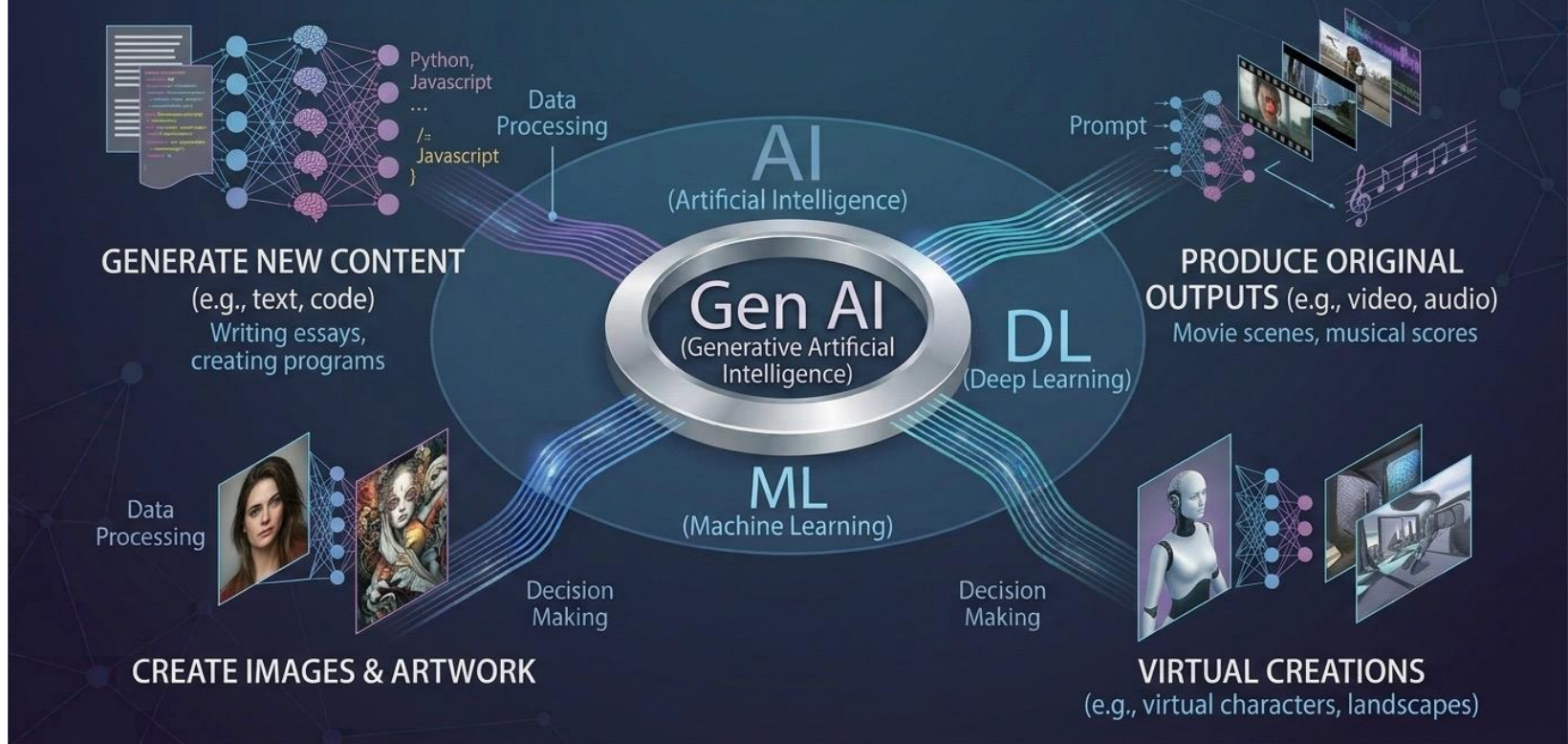


- **Deep Learning (DL)** is a **subset of Machine Learning (ML)** that uses **artificial neural networks** with multiple layers to learn from data.
- Inspired by the structure of the **human brain**, deep learning models automatically learn increasingly complex features from large amounts of data.

Why Deep Learning?

- Learns complex patterns with minimal manual feature engineering.
- Excels at processing **large-scale and unstructured data**.
- Delivers state-of-the-art performance on many AI tasks.

What is Generative AI?



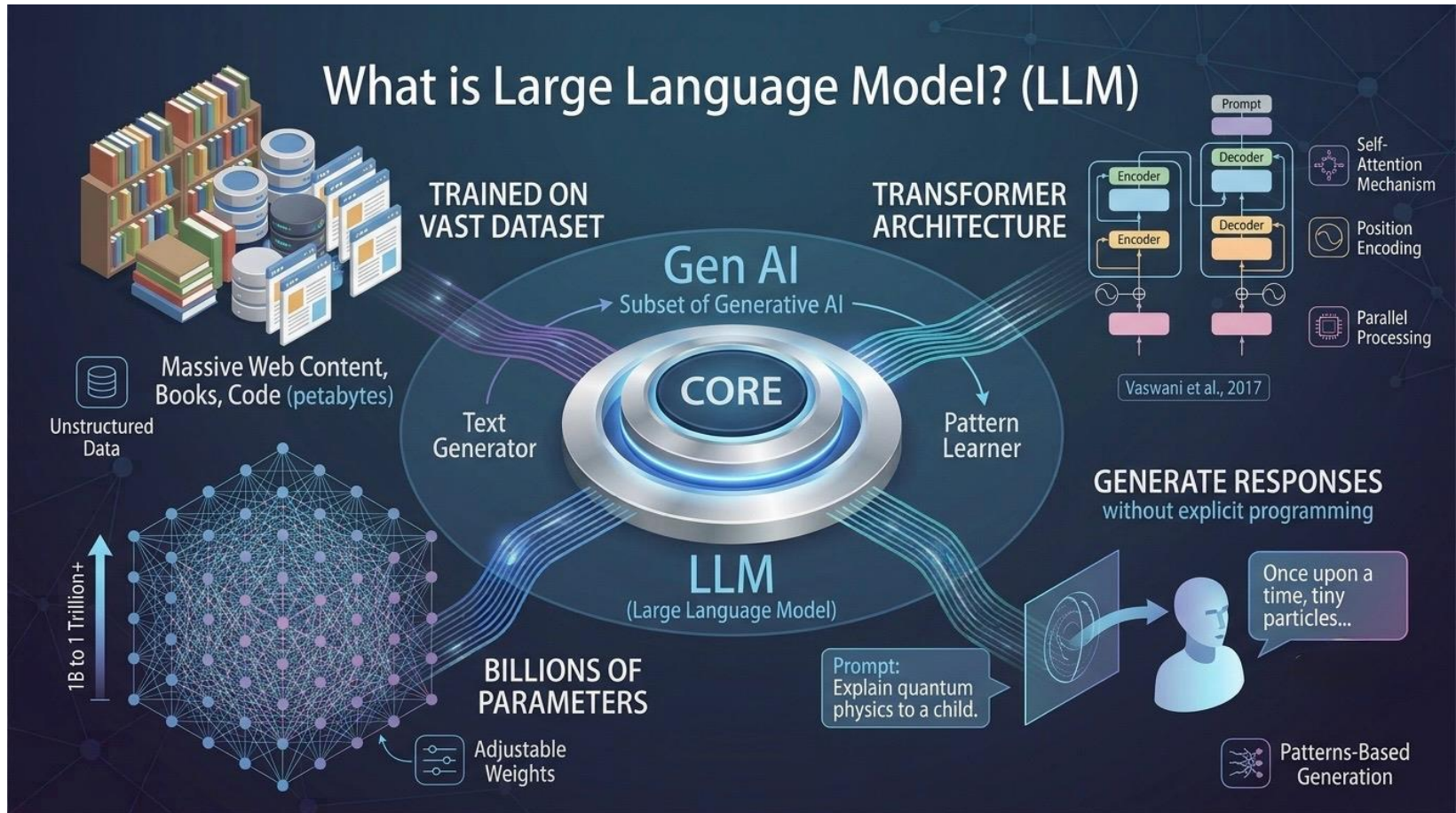
How is it Different?

Traditional AI / Machine Learning

- Focuses on **prediction, classification, and decision-making**.

Generative AI

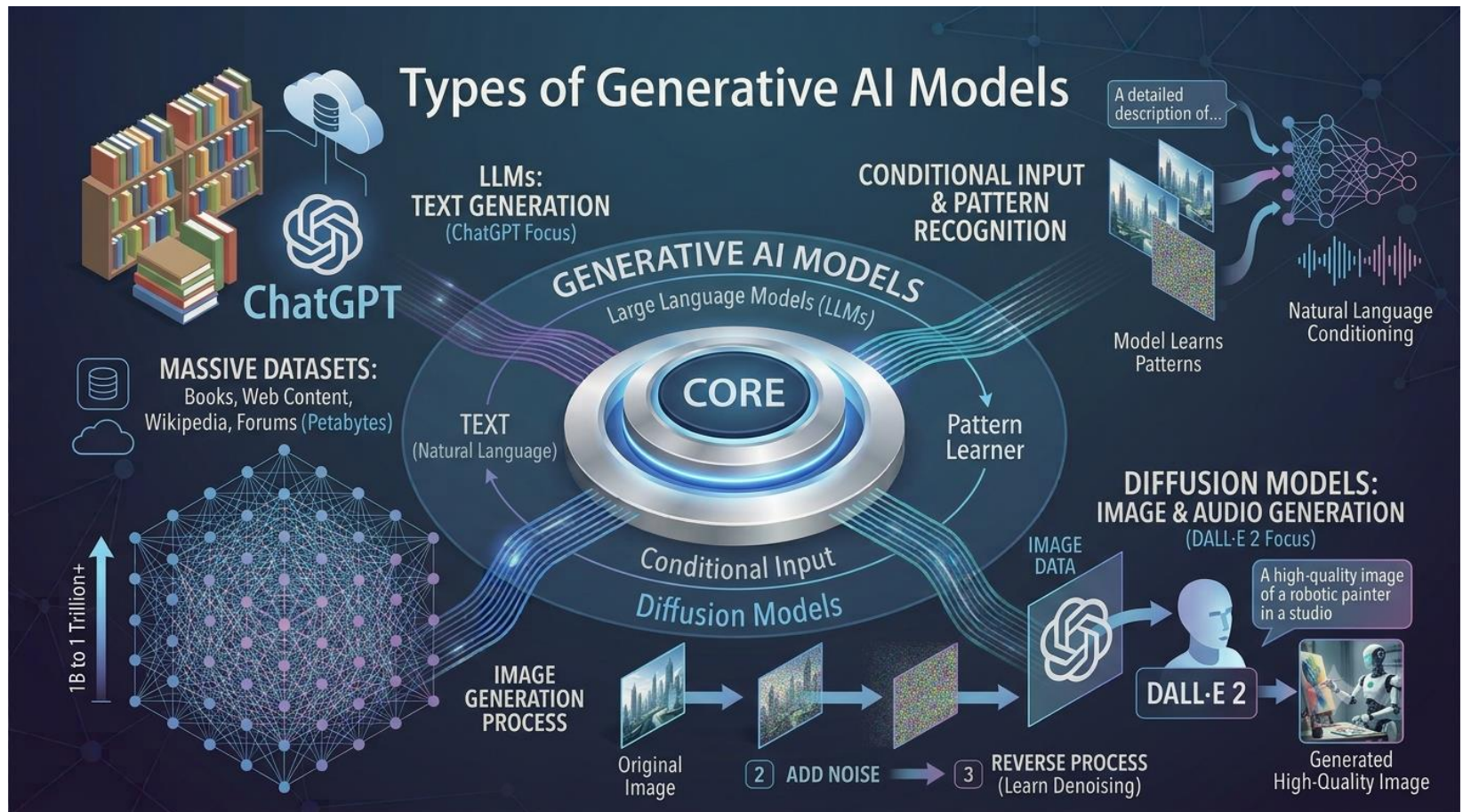
- Focuses on **creating original content** based on learned patterns.



- **Large Language Models (LLMs)** are a **subset of Generative AI** designed to **understand and generate human language**.
- They are trained on **massive amounts of text data** and contain **billions of parameters**, enabling them to learn complex language patterns.

How LLMs Work

- Built on the **Transformer architecture**.
- Learn statistical patterns and relationships in language from training data.
- Generate human-like text by predicting the **next most likely token (word or subword)** based on the given context.



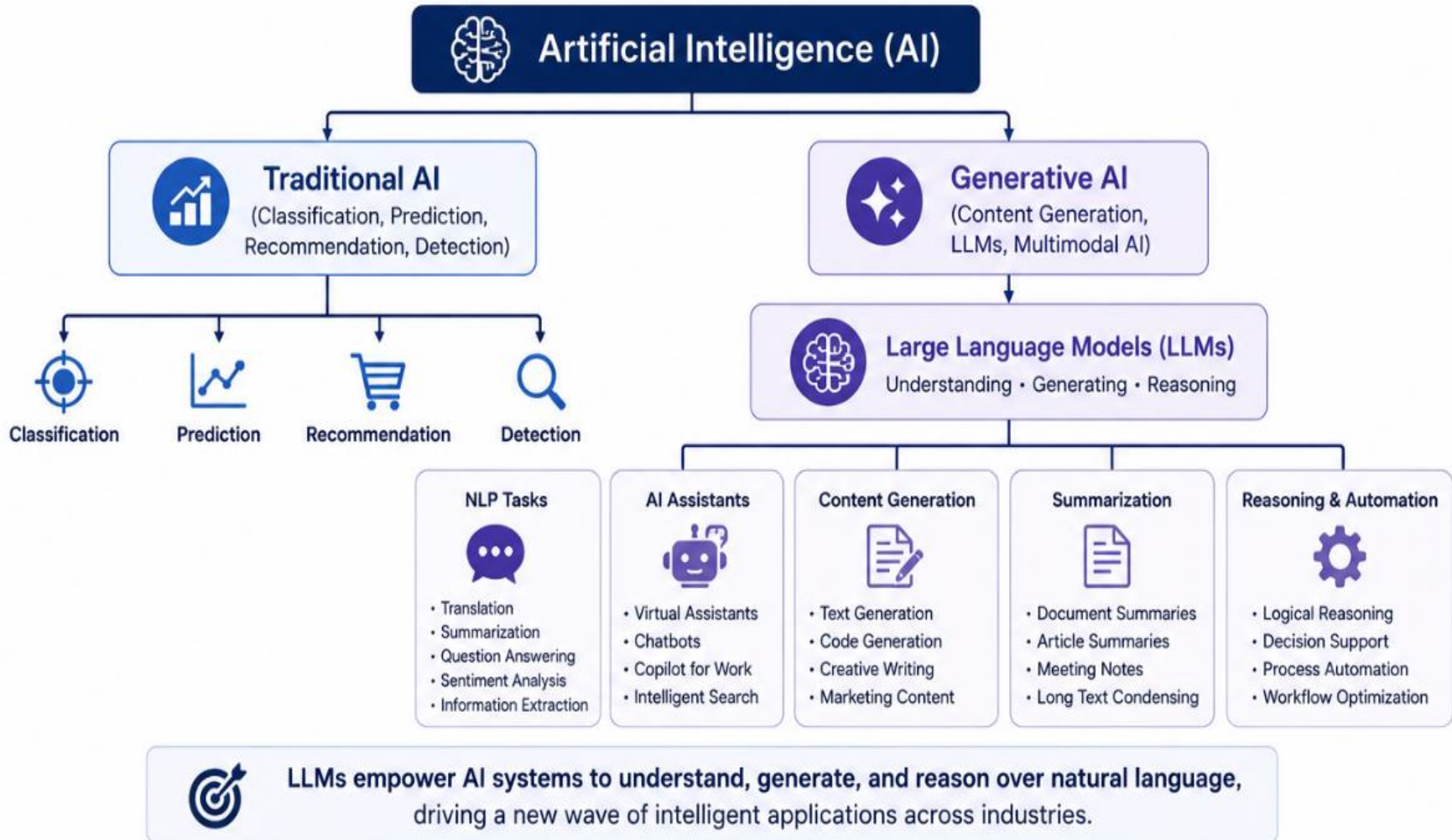
- Two of the most widely used Generative AI technologies are **Large Language Models** and **Diffusion Models**.
- LLMs specialize in understanding and generating text.
- ChatGPT is a well-known example of an LLM.

Diffusion Models

- Designed primarily for **image generation**, and can also generate **audio** and **video**.
- Generate content through a **denoising process**:
 - Start with **random noise**.
 - Gradually remove the noise over multiple steps.
 - Produce a realistic image. (e.g., DALL·E)

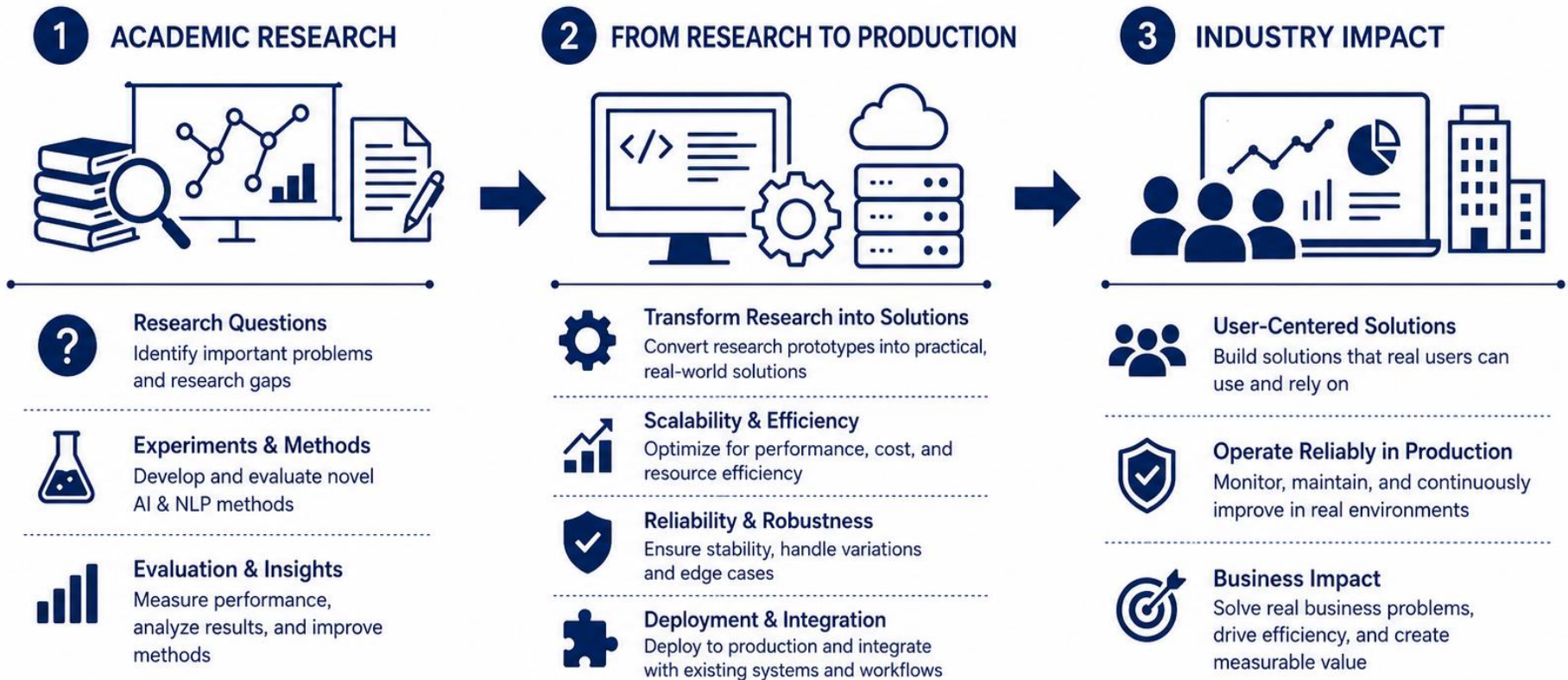
Overview of Current AI Technologies

From Traditional AI to Generative AI and LLMs



From Academic Research to Real-World Impact

Bridging research and industry to build production-ready AI & NLP solutions



Success is measured not only by model accuracy, but by real-world usability, reliability, and business value. This is how research knowledge creates real-world impact.

From Research to Production-Grade Translation



In Research

Our objective is to develop a high-quality machine translation model that produces accurate translations. However, even a strong NMT model may generate outputs with grammatical errors, inconsistent terminology, awkward phrasing, or translations that do not fully match the intended context.



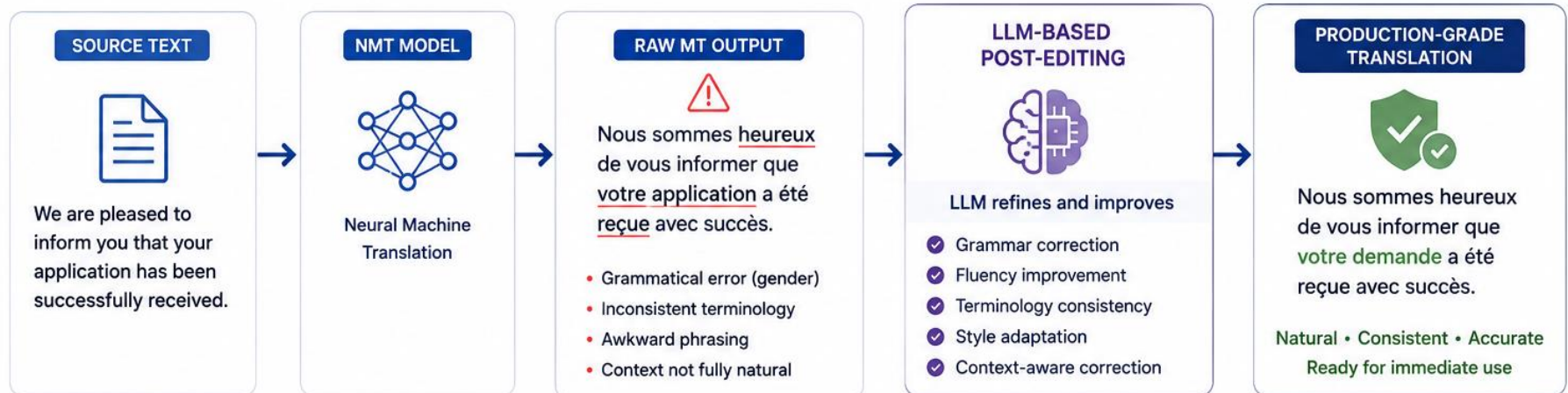
In Production

Customers expect translations that are natural, consistent, and ready for immediate use. To bridge this gap, we integrate a Large Language Model as an automatic post-editing component. Rather than retranslating the text, the LLM refines the machine translation by correcting grammatical issues, improving fluency, enforcing terminology consistency, and adapting the style while preserving the original meaning.



The Result

This combination leverages the strengths of both technologies: the NMT model performs efficient multilingual translation, while the LLM enhances linguistic quality. The result is a production-grade translation that requires significantly less human editing and provides a better user experience.



NMT + LLM
The Best of Both Worlds



NMT Strengths

- Fast and efficient translation
- Handles many languages
- Scalable for large volumes



LLM Strengths

- Deep language understanding
- Improves fluency and style
- Ensures consistency and correctness



Better User Experience

- Higher quality
- Less human editing
- Greater customer satisfaction

Career Advice & Growth in AI

- Build a strong foundation in **mathematics, programming, and machine learning**.
- Practice by **working on personal projects** and solving real-world problems.
- Learn to use modern AI tools and frameworks.
- Read research papers and stay updated with the latest AI trends.
- Participate in **AI competitions, hackathons, and internships** to gain practical experience.
- Share your work through **GitHub, a portfolio, or technical blogs**.
- Join AI communities, attend seminars, and learn from others.
- Be curious, keep experimenting, and **never stop learning**.



Thank You!